

## BAR, Montenegro

WMO: 134610

Lat: 42.100N

Lon: 19.100E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 34.8       | 37.8 | 6.4                           | 7.7 | 39.2 | 11.7 | 10.0 | 41.7 | 29.8                  | 34.1 | 24.9 | 39.0 | 11.4                  | 50   | 0.410 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 16.1                   | 89.8            | 75.9 | 88.2 | 75.7 | 86.8 | 75.0 | 79.6                | 87.6 | 78.2 | 86.5 | 76.9 | 85.3 | 8.5                  | 270  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 77.0                            | 140.8 | 86.8 | 75.5 | 133.4 | 85.5 | 73.9 | 126.5 | 84.4 | 43.1          | 87.9 | 41.7 | 86.3 | 40.3 | 85.4 | 84.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |    |    |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|----|----|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |    |    |
| (4) 20.8          | 18.1 | 15.9 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |
| (5)               |      |      | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 63.8   | 49.1  | 51.1  | 54.6  | 60.4  | 66.7  | 74.5  | 78.5  | 79.1  | 73.3  | 65.8  | 59.3  | 52.5 |
|                                                                                  | DBStd       | 11.33  | 5.81  | 5.39  | 4.44  | 3.61  | 4.55  | 4.46  | 3.26  | 3.05  | 4.13  | 3.97  | 5.25  | 5.69 |
|                                                                                  | HDD50       | 182    | 82    | 42    | 13    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 5     | 40   |
|                                                                                  | HDD65       | 1992   | 492   | 390   | 324   | 145   | 33    | 1     | 0     | 0     | 1     | 38    | 180   | 388  |
|                                                                                  | CDD50       | 5226   | 55    | 73    | 154   | 313   | 518   | 735   | 885   | 902   | 700   | 491   | 283   | 117  |
|                                                                                  | CDD65       | 1560   | 0     | 1     | 0     | 7     | 86    | 285   | 420   | 437   | 251   | 64    | 9     | 0    |
|                                                                                  | CDH74       | 13486  | 0     | 5     | 0     | 20    | 400   | 2389  | 4285  | 4491  | 1704  | 180   | 12    | 0    |
|                                                                                  | CDH80       | 4224   | 0     | 0     | 0     | 0     | 34    | 638   | 1523  | 1650  | 376   | 2     | 1     | 0    |
| Wind                                                                             | WSAvg       | 7.9    | 9.1   | 8.9   | 8.8   | 7.4   | 7.3   | 7.2   | 7.1   | 7.0   | 7.6   | 7.7   | 7.8   | 8.8  |
| Precipitation                                                                    | PrecAvg     | 55.2   | 5.8   | 5.3   | 5.0   | 5.1   | 3.0   | 2.3   | 1.2   | 1.8   | 5.5   | 6.0   | 7.7   | 6.9  |
|                                                                                  | PrecMax     | 77.0   | 13.8  | 13.2  | 14.2  | 11.6  | 6.6   | 9.1   | 3.9   | 7.5   | 14.1  | 9.9   | 17.0  | 13.5 |
|                                                                                  | PrecMin     | 39.9   | 1.0   | 0.6   | 0.0   | 0.7   | 0.4   | 0.2   | 0.0   | 0.4   | 0.2   | 2.8   | 0.0   | 0.0  |
|                                                                                  | PrecStd     | 10.9   | 3.7   | 2.8   | 3.3   | 2.6   | 1.7   | 2.0   | 1.1   | 2.0   | 3.7   | 2.8   | 2.9   | 3.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 64.3  | 70.4  | 69.8  | 76.8  | 83.6  | 89.8  | 91.9  | 92.5  | 87.2  | 79.6  | 74.6  | 67.8 |
|                                                                                  |             | MCWB   | 53.1  | 55.4  | 54.3  | 63.2  | 72.5  | 73.7  | 76.1  | 75.0  | 74.9  | 68.6  | 62.6  | 59.1 |
|                                                                                  | 2%          | DB     | 61.4  | 65.2  | 66.7  | 72.9  | 80.3  | 87.4  | 89.7  | 90.1  | 85.7  | 77.6  | 71.4  | 64.5 |
|                                                                                  |             | MCWB   | 55.4  | 54.0  | 55.5  | 60.9  | 68.9  | 74.9  | 76.4  | 76.2  | 73.9  | 68.0  | 63.0  | 57.8 |
|                                                                                  | 5%          | DB     | 59.7  | 61.4  | 64.5  | 70.4  | 78.1  | 85.7  | 88.1  | 88.6  | 84.1  | 76.1  | 69.8  | 62.8 |
|                                                                                  |             | MCWB   | 54.5  | 53.1  | 55.1  | 60.3  | 67.4  | 74.4  | 76.2  | 75.7  | 72.6  | 66.9  | 62.3  | 57.3 |
|                                                                                  | 10%         | DB     | 57.7  | 59.4  | 62.6  | 68.3  | 76.0  | 83.8  | 86.7  | 86.9  | 82.4  | 74.2  | 67.9  | 61.2 |
|                                                                                  |             | MCWB   | 52.3  | 52.6  | 54.3  | 59.2  | 66.2  | 72.8  | 75.4  | 74.9  | 71.3  | 65.4  | 60.5  | 55.3 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 60.0  | 60.2  | 60.1  | 66.2  | 74.6  | 79.4  | 81.6  | 81.6  | 78.3  | 72.3  | 67.6  | 62.4 |
|                                                                                  |             | MCDB   | 60.9  | 63.8  | 64.9  | 73.5  | 81.8  | 86.7  | 88.3  | 89.1  | 85.1  | 75.6  | 70.2  | 64.6 |
|                                                                                  | 2%          | WB     | 57.3  | 57.4  | 58.6  | 63.7  | 71.3  | 77.5  | 79.6  | 79.8  | 76.2  | 70.1  | 65.7  | 59.5 |
|                                                                                  |             | MCDB   | 59.6  | 60.6  | 63.2  | 70.4  | 78.9  | 85.6  | 87.6  | 88.1  | 84.1  | 75.7  | 69.8  | 62.9 |
|                                                                                  | 5%          | WB     | 54.9  | 55.9  | 57.3  | 62.0  | 69.1  | 75.8  | 78.2  | 78.3  | 74.2  | 68.4  | 63.7  | 58.1 |
|                                                                                  |             | MCDB   | 58.2  | 59.7  | 61.9  | 68.2  | 76.4  | 84.4  | 86.5  | 86.8  | 82.5  | 74.7  | 68.4  | 61.8 |
|                                                                                  | 10%         | WB     | 53.2  | 54.1  | 55.8  | 60.5  | 67.1  | 74.0  | 76.8  | 76.8  | 72.4  | 66.8  | 61.3  | 56.3 |
|                                                                                  |             | MCDB   | 56.9  | 58.8  | 60.5  | 66.4  | 74.4  | 82.5  | 85.4  | 85.3  | 81.0  | 72.7  | 66.1  | 60.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.5  | 12.4  | 12.9  | 14.2  | 15.2  | 15.4  | 16.0  | 16.1  | 15.4  | 15.1  | 12.8  | 12.0 |
|                                                                                  |             | MCDBR  | 12.1  | 13.8  | 15.2  | 16.9  | 17.0  | 16.4  | 16.8  | 16.7  | 16.6  | 17.6  | 14.2  | 12.8 |
|                                                                                  |             | MCWBR  | 8.4   | 8.8   | 11.0  | 11.8  | 12.2  | 12.8  | 13.6  | 13.4  | 12.9  | 12.3  | 9.3   | 9.4  |
|                                                                                  | 5% WB       | MCDBR  | 10.6  | 11.2  | 13.8  | 16.0  | 16.3  | 16.2  | 16.4  | 16.4  | 16.0  | 16.5  | 12.5  | 10.9 |
| MCWBR                                                                            |             | 8.5    | 8.6   | 11.2  | 12.0  | 12.4  | 13.4  | 14.2  | 14.3  | 13.5  | 12.3  | 9.7   | 9.0   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.338  | 0.361 | 0.404 | 0.454 | 0.442 | 0.425 | 0.426 | 0.438 | 0.419 | 0.391 | 0.365 | 0.340 |      |
|                                                                                  | taud        | 2.445  | 2.367 | 2.260 | 2.139 | 2.242 | 2.335 | 2.325 | 2.284 | 2.331 | 2.386 | 2.412 | 2.446 |      |
|                                                                                  | Ebn at Noon | 254    | 265   | 265   | 259   | 266   | 270   | 268   | 261   | 258   | 251   | 240   | 242   |      |
|                                                                                  | Edh at Noon | 25     | 31    | 39    | 47    | 43    | 40    | 40    | 40    | 36    | 30    | 25    | 23    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 534    | 775   | 1165  | 1534  | 1911  | 2195  | 2261  | 1998  | 1439  | 991   | 620   | 478   |      |
|                                                                                  | RadStd      | 59     | 115   | 156   | 157   | 146   | 135   | 111   | 117   | 109   | 96    | 61    | 84    |      |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (22 neighbors) | +0.77 | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | -140  | N/A   | N/A   |

Nomenclature: See separate page